# Vikrant Subramaniam

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## Education

**University of Maryland, College Park** College Park, MD
*Bachelor of Science in Computer Science* *May 2026*

**Relevant Coursework**: Object-Oriented Programming, Computer Systems, Data Structures and Algorithms, Advanced Data Structures, Database Design, Web Application Development, Algorithms for Data Science, Introduction to Machine Learning

**Extracurriculars**: UMCP Gaming (Web Development Team, Collegiate Player), UMD Chess Club

## Skills

**Languages**: C, C#, Python, JavaScript/TypeScript, Java, SQL, Kotlin, PHP

**Frameworks & Tools**: React, Node.js, MongoDB, MariaDB, PostgreSQL, Firebase, REST APIs, Docker, Git, GitLab, AWS, Linux/Unix

**Certifications**: CompTIA ITF+, IC3 Digital Literacy, Technical Support Fundamentals (Coursera)

## Experience

**UMCP Gaming** College Park, MD
*Software Engineer* *August 2023 – Present*

- Collaborated with a team of software engineers in an Agile/Scrum environment, resolving GitLab tickets for bug fixes, feature enhancements, and UI improvements to support a seamless user experience
- Implemented navigation features including dynamic season tabs, layout restructuring, and data filtering logic using React and JavaScript to improve usability and accessibility for team-specific statistics
- Developed full-stack features using C# for backend logic and React/JavaScript for frontend development, including designing and managing a MongoDB database to store team-specific season data (win/loss statistics and map veto/pick information) sourced from Faceit.com

## Projects

**Parking Finder App** | Kotlin, Firebase, Android SDK, Google Maps API

- Developed a full-stack Android application enabling users to locate, filter, and view real-time parking spots with Google Maps API integration
- Implemented AdMob interstitial ads and SharedPreferences to manage user preferences (e.g., price filters, dark mode, and favorite parking spots), improving user experience and application monetization
- Designed dynamic UI features including real-time price-based filtering, color-coded parking indicators, and MVC-based navigation between app views

**Supervised Learning Response Models** | Python, Pandas, Scikit-learn, NumPy

- Implemented Random Forest, K-Nearest Neighbors, and Support Vector Machine models to analyze student responses, focusing on factors such as demographics and technical exposure
- Performed feature scaling, dimensionality reduction with PCA, and correlation analysis to identify statistically significant relationships between response categories, achieving 90% accuracy and 50% precision using 10-fold cross-validation
- Evaluated model performance using confusion matrix, precision, and accuracy scores; visualized results with Seaborn and Matplotlib to identify trends

**Custom Terrain Mapping and Data Structure Implementation** | Java, TreeMap, TreeSet, LinkedList

- Designed and implemented a landscape mapping engine in Java to support real-time insertion, removal, and querying of non-uniformly distributed data points with elevation and depth tracking
- Created custom data structures using linked lists, TreeMap, and TreeSet to support efficient terrain management, achieving logarithmic time complexity and scaling up to 100,000 data points
- Designed dynamic update and query handling to maintain consistent performance across large datasets